

Zymo fecal experiment

TaxoViz maps, LRBinner bins and qPCR evidence

SMRL research report | 6 October 2026

The dominant groups are consistent with several added-culture candidates. The strongest unresolved findings are classifier conflict in bin 0, *E. coli*/lambda-related ambiguity in bins 7/11/13, and the association of map shape with read length. Version 2 changes the display; it has not demonstrated improved classification.

Scope: saved-run analysis of 290,227 intentionally selected early sequences, linked to all 9,674,223 reads in the LRBinner run. No labels were corrected and no embedding or model was rerun.

How to read the evidence

Observed: file checks, read joins, plotted coordinates and saved labels. Stored: notebook settings and historical logs. Hypotheses: biological and representation-based explanations for structure. Proposed: independent validation and correctness fixes. Tool agreement and bin membership are diagnostic evidence, not ground truth.

Contents

Research interpretation and reference comparison; original-coordinate figures for bins, taxa, length and GC; quantitative neighborhood diagnostics; additional findings and qPCR audit; complete bin inventory; methods, provenance and sources.

Companion: ZYMO_AGENT_CONTEXT_20261006.json contains the report text, complete aggregate evidence, all 23 bins, all 39 qPCR target records and taxonomic profiles. Large originals and read identifiers remain in the local ignored run archive.

Research interpretation

2026-10-05 · Descriptive review of saved outputs; no model rerun or taxonomy correction.

The maps recover several strong groups consistent with the added cultures, but also expose mixed bins, classifier disagreement, and strong read-length effects. **Version 2 is a different display, not demonstrated better classification.** All 290,227 plotted reads were matched by ID to the full LRBinner run. Neither the bin labels nor agreement colors are ground truth.

Data and reproducibility

The subset is the owner's **intentional earliest-sequence 3% selection using a custom ontime-based sampler**, not a random sample. It contains 2.958 Gbp, median length 5,159 bp, and mean GC about 42.9%. The full FASTQ contains 9,674,223 reads and 90.827 Gbp of sequence; its physical size is 184.19 GB / 171.55 GiB. Thus 3% refers to reads, not sequenced bases or file bytes. Full-run and early-run proportions answer different questions.

Both HTML versions contain identical read IDs, lengths, labels and agreement counts. The requested notebook, HTMLs, four SVGs, pickle and log are in the local run archive. The remote pickle is named `binning_result.pkl`, singular. All 12 downloaded run/source/support files matched remote SHA-256 checksums. The newly saved notebook is retained separately from the stale initial snapshot.

Setting	Version 1	Version 2
t-SNE perplexity	250	100
UMAP neighbors	100	50
UMAP min_dist	0.30	0.15
PCA provenance	Earlier fitted state not independently saved	43 PCs, 55.0% variance; cap 50

The SVG titles establish the three display settings. The current notebook records canonical k4+k6 CLR features, pseudocount 1, GC, Manhattan distance, seed 42 and t-SNE 500 optimization iterations plus early exaggeration. We cannot establish the complete version-1 fitted state from overwritten notebook cells. Old 5,730,402-read and 9,993-read statistics still survive in unrelated cells; they do not describe these plots.

What the structures mean

TaxoViz makes sequence features → PCA → t-SNE/UMAP, then colors reads by classifier agreement. It does not produce the LRBinner bins or correct labels. Agreement colors identify combinations of tools, not taxa. Genus/species panels share the same coordinates.

The bin overlay and taxon overlay use the original HTML coordinates with every plotted read retained. Bin numbers below refer to LRBinner; taxonomic percentages use agreement between KrakenUniq and Centrifuger (K=C) among **all plotted reads in that bin**, including unresolved reads in the denominator.

LRBinner bin	Full-run reads	Plotted reads	Main observation
4	2,562,702	74,268	Broad central/petal structure; 69.1% K=C <i>B. fragilis</i> , with other gut taxa mixed in
5	1,959,140	53,988	Large separate UMAP petal structure; 71.5% K=C <i>E. faecalis</i>
3	797,077	23,854	Well-separated island; 98.5% K=C <i>P. asaccharolytica</i>
1	556,436	18,476	Low-GC island; 89.0% K=C <i>F. nucleatum</i>

LRBinner bin	Full-run reads	Plotted reads	Main observation
2	416,364	11,466	High-GC island; 94.2% K=C <i>K. pneumoniae</i>
0	431,839	13,611	Coherent island, but only 2.5% K=C <i>P. anaerobius</i> ; strong tool disagreement
7	476,742	16,557	Isolated region with conflicting <i>E. coli</i> , lambda-phage and other labels

Bins 18/20/21/22 are mixed by these diagnostic labels. Bin 22 alone includes K=C *B. fragilis* (32.1%), *P. vulgatus* (8.2%) and *Blautia wexlerae* (5.5%). Conversely, one taxon can span bins: *P. asaccharolytica* also appears in bin 10, *K. pneumoniae* in bin 6, and *E. faecalis* in bin 18. This supports neither “one island = one species” nor “one bin = one species.”

The length/GC views show long reads along many petals/loops and shorter reads toward dense cores. GC separates major groups: bin 1 averages 27.0%, bin 5 37.6%, bin 3 52.9%, bin 2 58.4%. These are observations. Length-dependent sparse k6 counts and count-pseudocount CLR, local genomic composition, and nonlinear projection are plausible contributors to the petals; errors, strains, plasmids or distinct species are not established by shape. Quality was not available for this assessment.

Version 2 has tighter UMAP groups and a rearranged t-SNE map. For 10,000 fixed query reads and 15 nearest displayed neighbors, same-bin fractions are 89.0%→89.2% for t-SNE and 85.5%→86.4% for UMAP. However, exact neighbor overlap across versions is only 47.1% and 11.7%, respectively. These are **2D concordance diagnostics**, not accuracy, high-dimensional neighborhood preservation, or proof of improvement. UMAP's authors also caution against interpreting its apparent density and separations as original-space structure ([documentation](#)).

How LRBinner differs

The inspected run used k4 composition plus 15-mer coverage profiles, an 8-dimensional VAE, then iterative density-valley clustering using cosine-derived distances. The final resumed run used exhaustive search (-bit 0) and minimum bin size 500. It found 23 seed clusters, then assigned **1,448,297 remaining reads (15.0%)** by scores from composition/coverage profiles. Every full-run read consequently has a bin; this is not evidence that every read fits its bin confidently. Read-mode clustering here is not HDBSCAN on the TaxoViz plot. Source snapshot: LRBinner 23478e23fd7d74b26a87958cbf1efbcc273009de; see [method paper](#).

The pickle covers all 9,674,223 positions exactly once and matches bins.txt. The exact-ID join has zero missing/duplicate matched IDs, and all full FASTQ lengths match lengths.txt. LRBinner's coverage and fitted population are full-run quantities; TaxoViz's representation is fitted to the early subset. Their agreement is useful cross-method evidence, not an independent truth benchmark.

Zymo reference and qPCR

The [manufacturer's D6323 protocol](#) describes pooled human stool and links public DNA/16S characterization reports. The workbook's 354 reference percentages match the public 16S report's 354 values to six decimal places; named leading taxa match too. This strongly identifies the profile's provenance, but does **not** establish the physical sample's catalog number or lot. Reference profiles are method-dependent measurements, not exact read-level truth for the diluted, supplemented sample.

For illustration, the manufacturer's DNA reference has *Bacteroides* 18.44%, *Faecalibacterium* 8.09%, *Blautia* 6.17%, and *B. fragilis* 0.1033% of its reported classified profile. The workbook 16S profile instead has *Blautia wexlerae* 7.542%, *F. prausnitzii* 7.303%, and *B. fragilis* 0.0263%. In this early ONT subset, K=C *B. fragilis* accounts for 21.03% of all reads / 34.14% of bases; *E. faecalis* for 14.64% / 27.15%. The dominance of added-culture candidates is qualitatively consistent with the workbook, not a discrepancy against an unmodified mock standard.

The workbook records 20 µL Zymo, a 10-fold dilution, a 1000-fold culture-sample dilution, and a 6:1 culture:sample mixture. Those notes are the available record, not independently verified preparation details. Its culture list includes *F. nucleatum*, *E. faecalis*, *P. anaerobius*, *P. gingivalis*, *P. asaccharolytica*, PKS-labelled *Klebsiella*, and BFT-labelled *B. fragilis*.

qPCR cannot supply quantitative truth here. Sheet 2 omits the ×6 culture contribution for *E. faecalis* and *F. nucleatum* while applying it to several other targets. “PEP stom” conflicts with the *P. anaerobius* culture label; the PKS description names *E. coli* while the culture list names *Klebsiella*. BFT/PKS marker copies are not interchangeable with universal 16S copies or organism abundance. N/A is not zero; very late Cq results need assay controls/limits. Formulas and cached values are preserved in the qPCR audit.

Most useful follow-ups

- 1. Audit bin 0 and its classifier references.** KrakenUniq calls 13,368/13,611 reads *P. anaerobius*, MetaPhlAn calls 3,757, while Centrifuger spreads them across other taxa or leaves them unresolved. The culture record supports targeted validation, not automatic reassignment.
- 2. Inspect bins 7/11/13 for mobile/control sequence.** Bin 7 has 4,792 Centrifuger lambda calls and almost no MetaPhlAn species calls. Shared/mobile sequence, control material and database/parser effects are hypotheses requiring alignment evidence; do not label this island simply *E. coli* or delete it.
- 3. Fix agreement semantics before using colors for correction.** Of 43,794 species-level K_only reads, 37,289 have conflicting classified labels; 5,617/6,087 M_only reads also contain conflicts. “Only” currently hides disagreement. Genus-wide three-tool agreement is 20.94%, species-wide 11.07%; neither is calibrated confidence.
- 4. Freeze complete per-version configuration and fitted transforms, then test length/GC effects.** Retain the intentional early-time design and compare the same IDs. Assess changes against independent sequence evidence and rare/unknown retention, not visual compactness. P0 correctness tasks remain open.

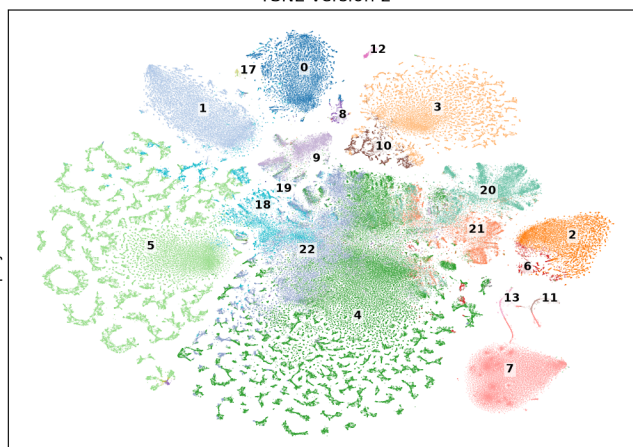
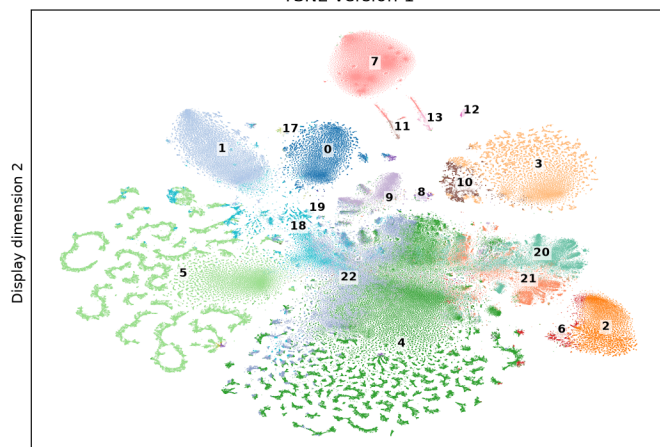
Detailed calculations, limits and provenance: experiment record. No biological accuracy, strain identity, toxin carriage, or correction gain was validated.

LRBinner bins on the TaxoViz maps

LRBinner full-run bins on TaxoViz coordinates | 290,227 identical reads

TSNE version 1

TSNE version 2

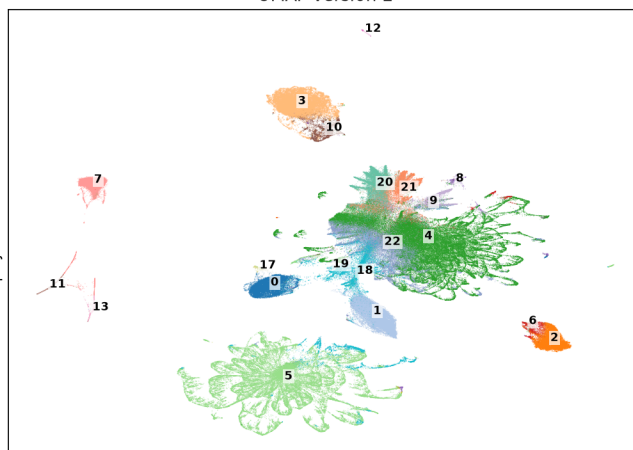
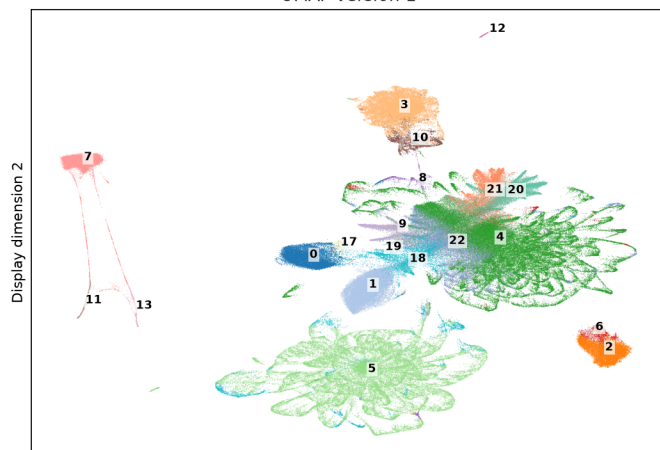


Display dimension 1

Display dimension 1

UMAP version 1

UMAP version 2

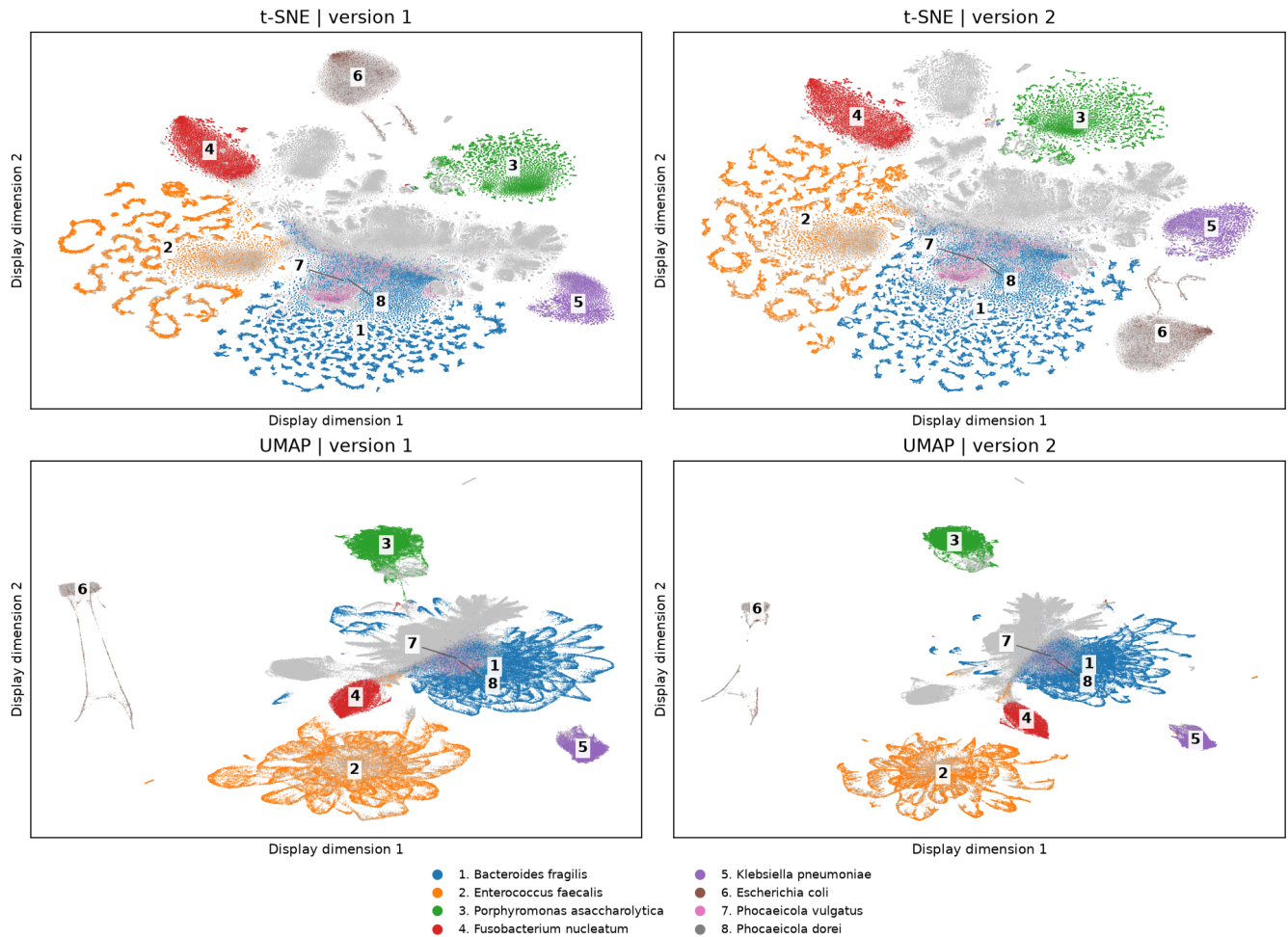


Display dimension 1

Display dimension 1

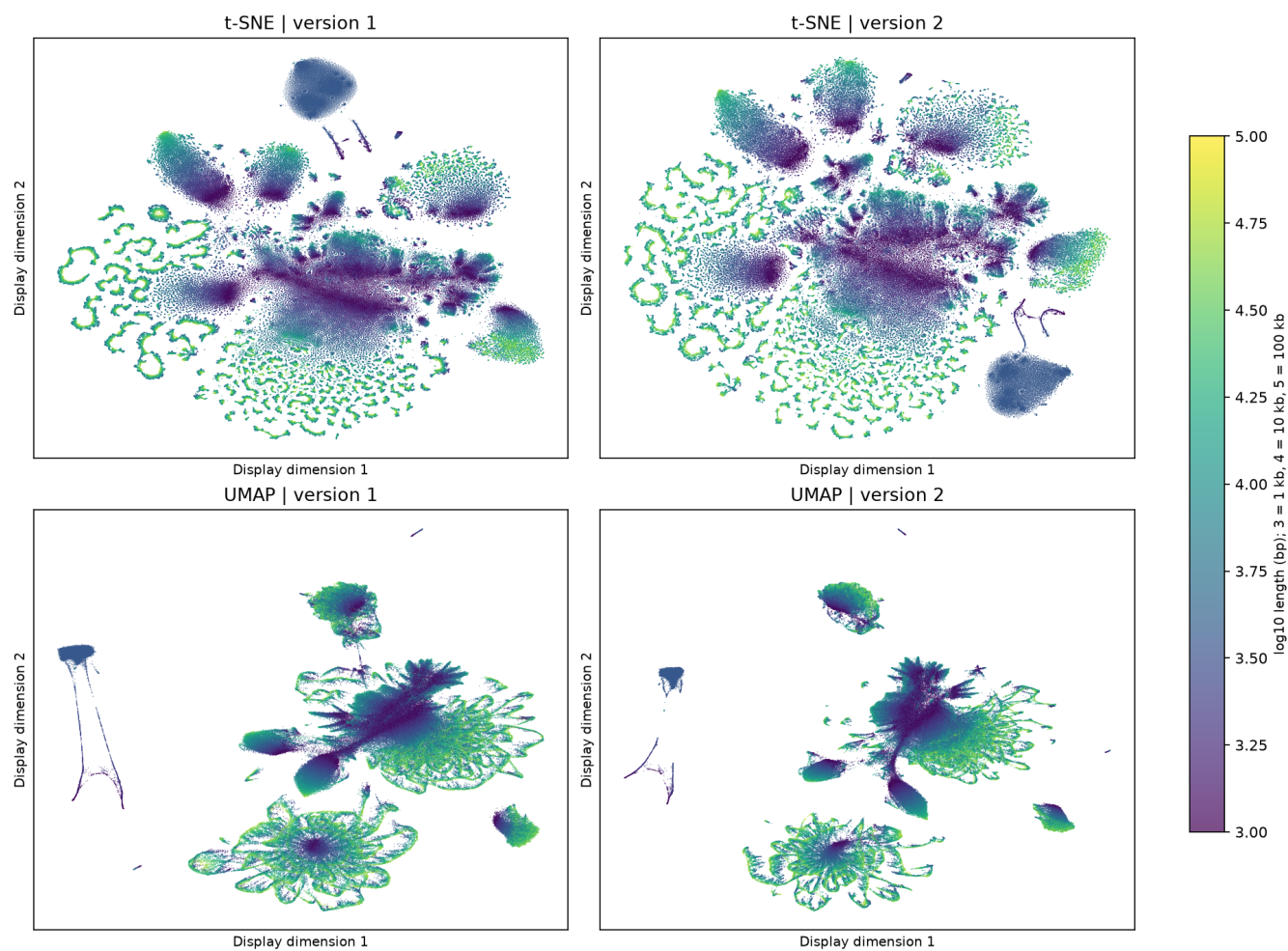
All 290,227 reads use the original HTML coordinates. Colors identify the 23 LRBinner bins. Small bins remain drawn even where text labels are suppressed. Several bins share broad regions, and bin membership does not establish species identity.

Taxonomic agreement on the same coordinates



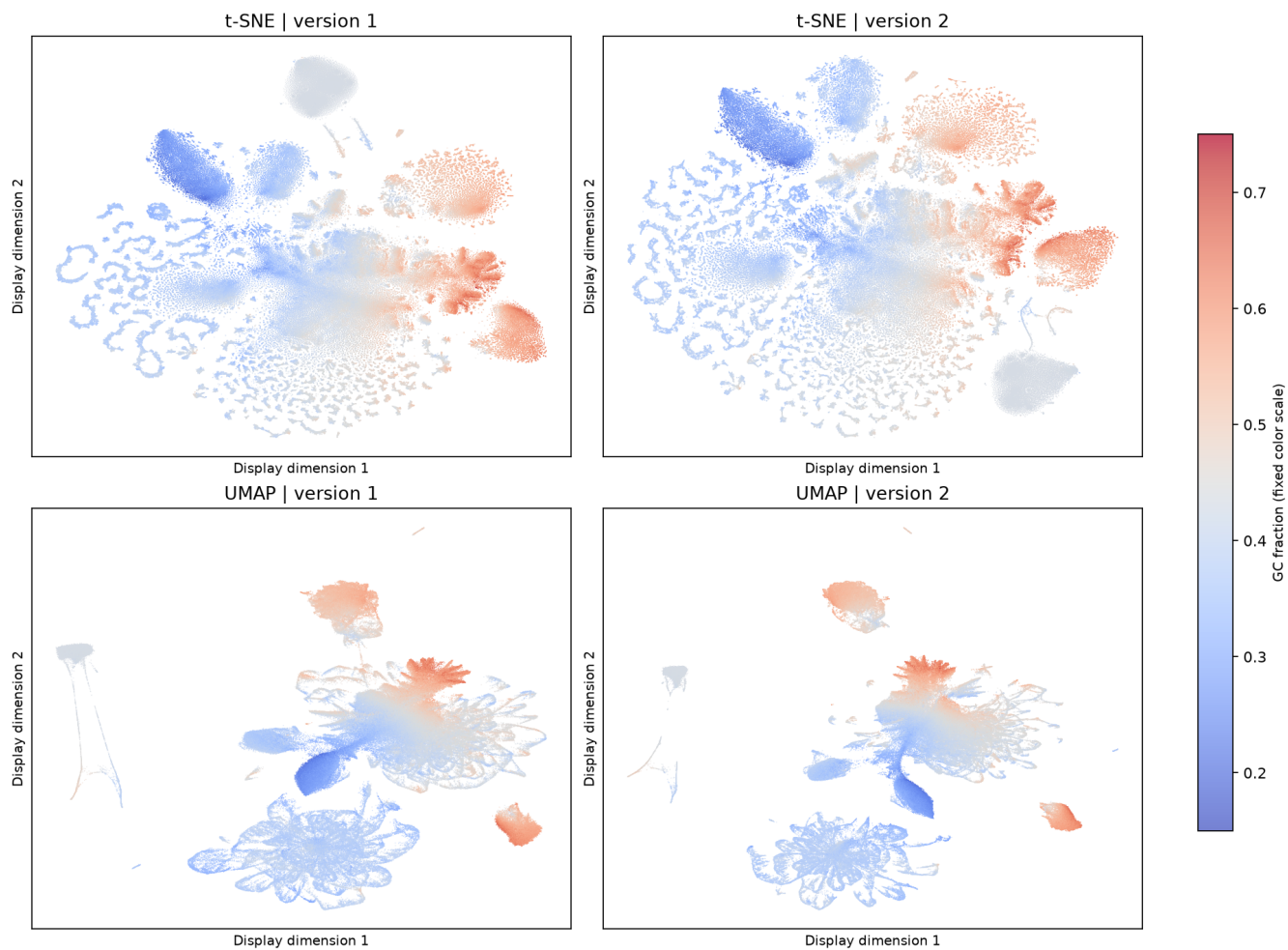
K=C denotes identical KrakenUniq and Centrifuger species strings, excluding unclassified. The leading eight species are colored; other resolved and unresolved calls remain visible. Numbers refer to the legend, not LRBinner bin IDs.

Read length and the petal structures



Shared log10 length scale across both methods and versions. Many peripheral petals contain longer reads, whereas dense cores often contain shorter reads. Shape alone cannot identify strains, plasmids or error classes.

GC composition across versions



Shared GC scale across both versions and methods. Major low- and high-GC islands recur, while geometry changes. GC is part of the TaxoViz feature input, so its organization is not independent validation.

Neighborhood diagnostics

Map	Same bin	Same K=C species*	Median log2 length ratio	Median GC difference
t-SNE v1	88.98%	90.46%	0.2497	0.00566
t-SNE v2	89.15%	90.02%	0.2342	0.00555
UMAP v1	85.54%	85.82%	0.3510	0.00928
UMAP v2	86.43%	86.39%	0.2886	0.00818

10,000 deterministic query reads, 15 exact Euclidean neighbors in each saved 2D map, seed 42; self excluded. *Only pairs resolved at both ends contribute to species agreement: 90,723 / 90,448 / 88,183 / 88,480 pairs respectively. Other diagnostics use all 150,000 directed pairs.

Exact cross-version neighbor overlap is 47.13% for t-SNE and 11.65% for UMAP. Cross-method overlap is 15.99% in v1 and 19.20% in v2. Broad concordance can coexist with substantially changed fine neighborhoods. No confidence interval, accuracy claim or original-space preservation test is implied.

Length interval (bp)	Reads	K=C species resolved
0 < L <= 2,000	53,674	52.59%
2,000 < L <= 5,000	88,754	55.28%
5,000 < L <= 10,000	54,747	70.39%
10,000 < L <= 30,000	70,061	86.91%
30,000 < L <= 100,000	22,991	97.06%

Legacy agreement categories hide conflicts: 37,289 of 43,794 species K_only reads and 5,617 of 6,087 M_only reads contain conflicting classified calls. Even 29,310 of 166,911 KC-category species reads have a conflicting third-tool label. Three-tool agreement is 20.94% at genus and 11.07% at species; it is not calibrated confidence.

Additional findings and qPCR

Additional findings and interpretation

Length is associated with classifier agreement: K=C species resolution rises from 52.6% for reads up to 2 kb to 97.1% above 30 kb. This is not a measured error rate or a causal length effect; taxon and length distributions are confounded. The independent length and GC panels help distinguish these associations from taxonomic islands.

K=C species labels resolve 199,035 reads (68.58%); 91,192 reads remain unresolved (31.42% of reads, 15.17% of bases). *B. fragilis* and *E. faecalis* together account for 35.67% of reads but 61.29% of bases. Read counts and base counts therefore produce substantially different pictures of dominance. Neither equals cell abundance.

P. gingivalis is detected sparsely: 42 MetaPhlAn, 407 KrakenUniq and 406 Centrifuger calls. *Gemella*-related genus calls are also sparse (22, 2 and 19 respectively); Centrifuger has only two *G. morbillorum* species calls. These are per-tool observations, not confirmed counts of true organisms, and sparse calls do not establish absence relative to the workbook.

Bin 0 illustrates why a coherent island cannot resolve conflicting taxonomy. KrakenUniq labels 13,368 of 13,611 reads *P. anaerobius* (98.2%), MetaPhlAn 3,757, and Centrifuger only 338. Centrifuger additionally assigns *P. equinus* (1,262), *Caudoviricetes* (1,078), *P. porci* (955), or unclassified (3,276). Independent sequence evidence is needed before reassignment.

Bin 7 contains 16,557 reads: Centrifuger assigns *E. coli* to 8,674 and lambda phage to 4,792, whereas KrakenUniq assigns *E. coli* to 8,105 and MetaPhlAn leaves 16,553 unclassified at species level. Related ambiguous bins 11 and 13 contain 981 and 1,028 reads. Bin 11 includes 557 Centrifuger lambda calls; bin 13 includes 678 *E. coli* calls. Mobile sequence, controls, reference overlap and parser effects remain competing hypotheses, not demonstrated contamination.

LRBinner's earlier attempts yielded 8, 15 and 19 bins before the final 23-bin result. The final resumed invocation took 52,249.43 seconds (14.5 hours), excluding preprocessing and training. Exhaustive cluster searching followed by assignment of remaining reads explains complete bin coverage; coverage alone does not measure bin purity.

qPCR audit detail

The pure-Zymo block reports universal 16S Cq 19.87 / 1,400,000 copies, BFT 39.88 / 1.59, Porphyr 39.04 / 2.8, PKS 38.12 / 7.98 and Gem 36.67 / 25.1. The separate culture-associated block reports 16S 18.60 / 3,302,784, BFT 20.64 / 716,909, ENTRC 20.09 / 1,393,727, Fus 24.44 / 89,731, Gem 26.55 / 16,500, Pep 23.07 / 286,828, PKS 31.87 / 578, Porphyr 21.72 / 256,648 and Strep 24.27 / 20,188. These are workbook values; the second block is not explicitly identified as the sequenced mixture.

The common Sheet 2 formula combines $H10/5 + 60 \cdot 1000/5$, but ENTRC and Fus omit the factor 6. Preparation notes do not establish which convention is correct. All 39 target formulas, cached results, descriptions and cell locations are preserved in the JSON handoff. They were read, not recalculated or silently corrected. No toxin carriage, strain identity or absolute organism concentration is validated.

Complete 23-bin inventory

Ordered by plotted size. Taxon percentage uses every plotted read in the bin, including unresolved reads. Selection % is plotted/full-bin reads. Top taxon is K=C species agreement, not a truth or purity estimate.

Bin	Full reads	Plotted	Selection %	Top K=C species	% of bin
4	2,562,702	74,268	2.90	Bacteroides fragilis	69.1
5	1,959,140	53,988	2.76	Enterococcus faecalis	71.5
3	797,077	23,854	2.99	Porphyromonas asaccharolytica	98.5
22	722,921	22,033	3.05	Bacteroides fragilis	32.1
1	556,436	18,476	3.32	Fusobacterium nucleatum	89.0
7	476,742	16,557	3.47	Escherichia coli	27.0
0	431,839	13,611	3.15	Peptostreptococcus anaerobius	2.5
20	417,779	14,423	3.45	Faecalibacterium prausnitzii	13.4
2	416,364	11,466	2.75	Klebsiella pneumoniae	94.2
18	397,514	11,714	2.95	Enterococcus faecalis	29.0
21	287,107	9,901	3.45	Alistipes shahii	4.9
9	239,659	8,505	3.55	Phocaeicola vulgatus	9.6
10	137,826	3,848	2.79	Porphyromonas asaccharolytica	38.0
6	97,913	2,525	2.58	Klebsiella pneumoniae	50.3
8	52,248	1,550	2.97	Peptostreptococcus anaerobius	19.5
13	40,275	1,028	2.55	Escherichia coli	26.4
11	32,931	981	2.98	Escherichia coli	14.9
12	17,300	639	3.69	none	0.0
17	13,323	399	2.99	Fenollaria massiliensis	6.3
19	6,825	200	2.93	Streptococcus pyogenes	9.5
14	5,061	124	2.45	Bacteroides fragilis	75.0
16	4,215	108	2.56	Enterococcus faecalis	69.4
15	1,026	29	2.83	Porphyromonas asaccharolytica	100.0

Methods, provenance and sources

Status: completed descriptive review. P0 fixes and biological validation remain open.

Owner: SMRL research workspace; requested by owner on 2026-10-05.

Hypothesis: broad sequence structure agrees with some added-culture evidence, while local map structure may reflect read length, GC, or ambiguous sequence.

Decision to inform: which groups need targeted validation, and whether version 2 establishes scientific improvement (it does not).

Provenance and scope

- Local baseline/workspace HEAD: 2e29b4b9f6b15631d2c588fbff708148efc1de6f; production baseline previously audited: 9528b62e5da4624384c9977561a6e8caa484702a.
- Focused branch: research/zymo-fecal-review-2026-10-05. Pre-existing working-tree changes affected 24 files but disappeared under git diff --ignore-space-at-eol; preserved without normalization. At the close of the October 5 analysis, no commit, merge, push, production-code edit or remote training had been performed. Report publication and Git synchronization are a separate October 6 delivery step.
- Connection: existing WSL SSH alias used; private-key mode was tightened from 0755 to 0600 so OpenSSH would accept it. No key material was read into the report or copied.
- Remote source: dgx03-ngate:/mnt/raid0/Toxins/test/lrbinner/zymo_fecal_run.
- LRBinner source HEAD: 23478e23fd7d74b26a87958cbf1efbcc273009de. Remote status showed modified build.sh and untracked auxiliary/hmmer-3.4/; relevant Python files were archived separately. The source snapshot describes current source, not proof that every historical invocation used an identical unmodified environment.
- Root notebook was stale on first retrieval. Owner saved the active session, after which the updated notebook was fetched and verified. Both snapshots are retained. The original notebook hash is 5c7e5127ceef24a6c93987e3b0d35a5b520bf4a3d5fc66efd4ff540315255d5e; updated hash is in the artifact manifest.
- Local artifacts: runs/zymo_fecal_20261005/ (already ignored by Git). This includes all requested files, exact server TaxoViz core, qPCR workbook copy, LRBinner source snapshot, bins.txt/lengths.txt, public manufacturer reports, per-read joins and diagnostic figures. Original root workspace code/notebook remain untouched.
- Compact evidence suitable for Git: evidence directory. Raw read IDs, large HTML/notebook/pickle files and copied upstream code remain outside Git under runs/.

Configuration, inputs and environment

Saved notebook: k=(4,6), CLR count pseudocount=1; PCA cap=50, target=0.55, stored result=43 PCs/55.0%; t-SNE perplexity=100, metric=Manhattan, n_iter=500, early-exaggeration output=250 iterations; UMAP neighbors=50, min_dist=0.15, Manhattan, 200 epochs in stored output; seed=42. SVG v1 titles give perplexity=250, neighbors=100, min_dist=0.3. Complete v1 PCA configuration/fitted transform is not independently retained. Parameter-only attribution is therefore limited.

Stored notebook environment (not newly verified on server): numpy 2.2.6, pandas 2.3.3, matplotlib 3.10.9, openTSNE 1.0.4, umap 0.5.12, sklearn 1.6.1, numba 0.65.1, joblib 1.5.3. MetaPhlAn DB path identifies mpa_vJan25_CHOCOPhAnSGB_202503.pkl. Actual classifier program/database hashes, taxonomy hashes/versions, fitted models and complete seeds for the historic LRBinner run were not supplied/verified. Do

not infer them from filenames. Input/output artifact hashes are in artifact_manifest.json.

User-confirmed sampling: custom ontime-based earliest-sequence 3% selection. Timing fields/sampler code were not audited. Matching IDs lie at zero-based full-file positions 48,856–1,040,930; file order is not assumed to equal acquisition-time order. This is documented intentional selection, not a sampling failure. A separate subsample/seqkit_sample dataset exists remotely and was not substituted into this review.

Full input FASTQ: 184,192,836,119 bytes; 9,674,223 reads; 90,826,618,124 nucleotide bases. Plot input: 290,227 reads; 2,957,983,218 bases. Full FASTQ content hash was not computed; size/mtime were recorded during extraction. Its full sequences were not copied locally.

Executed checks and descriptive analysis

1. Verify 12 transferred run/source/support files against remote SHA-256. Workbook and reference-download hashes also recorded.
2. Parse Plotly JSON directly, without executing embedded JavaScript. Decode typed binary coordinate arrays. Assert unique read IDs, identical rank/method row sets, identical coordinates between genus/species panels, and identical IDs/metadata across versions.
3. Perform one read-only remote FASTQ scan; validate every four-line record, plus marker and sequence/quality length; compare every length to LRBinner lengths.txt; preserve original row positions. Zero missing plotted IDs and zero duplicate matched IDs. Global duplicates among nonmatched IDs were not tested. Extraction took about 225.5 seconds; no inference was run.
4. Load pickle with globals forbidden in the reproducible audit; verify 23 bins cover every full-read position exactly once and match bins.txt exactly. Join by ID and source position; plot lengths and rounded GC agree with directly measured sequences.
5. Compute label counts, read fractions and nucleotide fractions separately. K=C is equality of stored rank-specific name strings, with unclassified excluded, and is explicitly diagnostic evidence. No truth labels or taxonomic accuracy measures are inferred.
6. For 10,000 deterministic query reads (numpy generator seed 42), compute exact Euclidean k=15 neighborhoods in each saved 2D map using scipy cKDTree; exclude self. Report same-bin fraction, same-K=C species fraction among pairs resolved at both ends, and cross-version neighbor overlap. Metrics do not evaluate PCA-space fidelity or classifier accuracy; no threshold was tuned and no confidence intervals/test claim made.
7. Render all 290,227 points on the original coordinates, with deterministic shuffled drawing order, to inspect full-bin and K=C overlays. Inspect source gradient raster panels and generated overlays. Plot coordinates were not refitted. Small bins (<150 plotted reads) remain drawn but are not text-labelled. The October 6 publication separates the nearby taxon annotations with leader lines; CSVs preserve exact identities.
8. Read workbook formulas and cached values without modifying/recalculating the original. Manufacturer 16S report and workbook have identical multisets of 354 percentage values after rounding to six decimals; leading named taxa also match. This is source-provenance evidence, not proof of the physical lot.

Local analysis used an isolated ignored Python 3.11 environment; exact package list is in analysis_environment.txt. Initial parsing used the available bundled Python 3.12/pandas/numpy. CPU-only analysis; GPU not used. Peak RSS, local CPU time and total temporary disk peak were not measured. Historical log records final LRBinner resumed invocation duration 52,249.43 s, which excludes earlier preprocessing/training and failed/repeated runs; do not report it as end-to-end runtime. Historical VAE: 8 latent dimensions, hidden [128,128], 200 epochs; first invocation reported CUDA absent, later resumed invocation CUDA present.

Results, limitations and next action

Research report, plot statistics, bin table, comparison diagnostics.

There was no train/development/test split, truth scoring, correction, relabelling or heavy benchmark. Correct→wrong / wrong→correct / classified→abstained counts: not applicable. Rare/unknown/mobile signals remain unresolved; all 23 bins are represented in the plotted subset (29–74,268 reads/bin). Negative finding: qPCR formulas/labels do not support an exact mixture truth; changed display neighborhoods do not establish v2 superiority. Species name ambiguities, old taxonomy synonyms and non-independent classifier errors remain.

Next: TV-001/002/003/004/007 correctness work, plus targeted independent alignment of bins 0 and 7/11/13, and length/GC-controlled representation experiments after provenance is frozen. These are proposed next tasks, not work executed here. See ADR-006 in DECISIONS.

Reproduction and source links

Local scripts under scripts/research/: analyze_zymo_plots_20261005.py, compare_zymo_lrb_20261005.py, audit_zymo_sources_20261005.py; execute with the isolated run environment. map_zymo_reads_20261005.py runs under WSL and requires the authorized SSH alias; it performs the remote read-only scan. It writes no remote output files. Run the plot parser, mapping, comparison, then source audit. No upstream production modules are imported/executed.

Public sources accessed 2026-10-05:

- [Zymo D6323 protocol](#), Appendix A: links and methods.
- [Zymo DNA reference report ZIP](#); extracted All/AbundanceTables/3.Genus/abun_table.tsv and 4.Species/abun_table.tsv. Fractions are within the report's classified profile, not all physical organisms.
- [Zymo 16S report ZIP](#); Composition_Barplots/Composition_Summary_L7.txt and corresponding heatmap table.
- [Characterization metadata workbook](#).
- [Zymo Fecal Reference Database](#); download interface required login. Public Appendix A downloads were used instead; no login bypass.
- [LRBinner paper](#), [source repository](#), [UMAP authors' clustering caveats](#).